AMENDED CLAIMS

1. A computer-implemented device (900) for processing a two-agent system input to form multiple at least partially optimised outputs, each output indicative of an action policy for each of the two agents, the device comprising one or more processors (901) configured to perform the steps of:

receiving (601) the two-agent system input, the two-agent system input comprising a definition of a two-agent system and defining behaviour patterns of two agents based on system states;

receiving (602) an indication of an input system state; and

performing (603) an iterative machine learning process (800) to estimate multiple aggregate functions, each representing the behaviour patterns of the two agents over a set of system states, wherein the multiple aggregate functions are determined in a single iteration of the iterative machine learning process,

wherein the multiple aggregate functions correspond to multiple best-response policies for the agents,

wherein the two-agent system is a two-player zero-sum game,

wherein the step of performing an iterative machine learning process comprises repeatedly performing the following steps until a predetermined level of convergence is reached:

generating (801) a set of random system states;

estimating (802) based on the two-agent system input the behaviour patterns of the two agents in the system states;

estimating (803) an error between the estimated behaviour patterns and the behaviour patterns predicted by each of multiple predetermined candidate aggregate functions, the error representing the level of convergence; and

adapting (804) the multiple predetermined candidate aggregate functions based on the estimated behaviour patterns, and

wherein the set of random system states are generated based on a predetermined probability distribution.

2. A device (900) as claimed in claim 1, wherein the processor (901) is configured to iteratively process the multiple aggregate functions for the input system state to estimate multiple at least partially optimised set of actions for each of the two agents in the input system state.

3. A device (900) as claimed in claim 1 or claim 2, wherein the multiple aggregate functions are determined in a single iteration of the iterative machine learning process in parallel.

4. A device (900) as claimed in any preceding claim, wherein multiple aggregate functions are determined in each iteration of the machine learning process.

5. A device (900) as claimed in any preceding claim, wherein the iterative machine learning process (800) is so as to promote behavioural diversity among the multiple aggregate functions determined in each iteration.

6. A device (900) as claimed in any preceding claim, wherein the iterative machine learning process (800) is performed in dependence on a diversity measure, wherein the diversity measure is modelled by a determinantal point process.

7. A device (900) as claimed in any preceding claim, wherein the multiple at least partially optimised outputs each comprise a collectively optimal action policy for each of the two agents in the input system state.

8. A device (900) as claimed in any preceding claim, wherein the multiple at least partially optimised outputs each represent a Nash equilibrium behaviour pattern of the two agents in the input system state.

9. A device (900) as claimed in any preceding claim, wherein the agents are autonomous vehicles and the system states are vehicular system states.

10. A device (900) as claimed in any of claims 1 to 8, wherein the agents are data processing devices and the system states are computation tasks.

11. A method (600) for processing a two-agent system input to form a multiple at least partially optimised outputs each indicative of an action policy, the method comprising the steps of:

receiving (601) the two-agent system input, the two-agent system input comprising a definition of a two-agent system and defining behaviour patterns of two agents based on system states;

receiving (602) an indication of an input system state; and

performing (603) an iterative machine learning process to estimate multiple aggregate functions, each representing the behaviour patterns of the two agents over a set of system states, wherein the multiple aggregate functions are determined in a single iteration of the iterative machine learning process,

wherein the step of performing an iterative machine learning process comprises repeatedly performing the following steps until a predetermined level of convergence is reached:

generating (801) a set of random system states;

estimating (802) based on the two-agent system input the behaviour patterns of the two agents in the system states;

estimating (803) an error between the estimated behaviour patterns and the behaviour patterns predicted by each of multiple predetermined candidate aggregate functions, the error representing the level of convergence; and

adapting (804) the multiple predetermined candidate aggregate functions based on the estimated behaviour patterns, and

wherein the set of random system states are generated based on a predetermined probability distribution.

12. The method (600) of claim 11, further comprising the step of causing each of the agents to implement a respective action of the at least partially optimised set of actions.

13. A computer readable medium (902) storing in non-transient form a set of instructions for causing one or more processors to perform the method (600) of claim 11 or 12.